

Your 12-Month Strategic Execution Plan to ₹70L+ (Senior Data Engineer → AI/Data Architect)

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Executive Summary

You are at an inflection point. With **13+ years of experience and 8+ years in data engineering**, combined with your **acknowledged strengths in continuous learning and AI interest**, you have the foundational platform to command **₹70L+ compensation** in India's booming data/AI market. However, the gap is not experience—it's **strategic positioning** around **AI/ML, LLMOps, and architectural leadership**.

This plan maps a **12-month roadmap** that bridges that gap through **project-driven learning, visible contributions, and market-ready positioning**. The core strategy: build **end-to-end AI/LLM projects on your existing data pipeline expertise**, secure relevant certifications, and reposition your profile as a **Cloud Data Architect + AI Integrator**—a role now commanding **₹60-90L+ packages** from top-tier companies and remote-first organizations.

Part 1: Current Market Position & Opportunity

Your Strengths

✔ Solid Foundation (Already 60-70% there):

- 13+ years total experience with 8+ in data engineering
- Expert-level proficiency in big data: Spark, Hadoop, Hive, Kafka
- Cloud platforms: GCP (BigQuery, Dataflow), Snowflake, Databricks
- ETL/pipeline optimization, automation (Airflow, Jenkins)
- Real-time streaming and batch processing at scale
- Mentorship & leadership experience (cross-functional collaboration)

Market Alignment: These skills align with **₹20-40L baseline** for Senior Data Engineers in India. Remote USD-equivalent roles offer **₹45-60L**.

Critical Gaps (The ₹30-50L Bridge)

✗ Why You're Not at ₹70L Yet:

1. **No AI/ML Model Experience** – The industry pivot of 2024-2025 now expects data engineers to understand ML pipelines, feature engineering, and model deployment.
2. **Limited LLM/GenAI Hands-on Work** – Generative AI is the differentiator. Candidates with LLM integration projects command **20-40% salary premium**.
3. **No MLOps/LLMOps Specialization** – Tools like Kubeflow, MLflow, LangSmith are now expected; you have exposure but no production track record.
4. **Missing Cloud Certifications** – Formal GCP/AWS Data Engineer certs validate skills and unlock tier-2 company roles (₹50-70L).
5. **Limited Public/GitHub Visibility** – Hiring managers now expect to see GitHub projects or LinkedIn portfolio; yours likely doesn't showcase AI/data work.

Market Truth: A Senior Data Engineer *without* AI/ML know-how tops out at ₹50-60L even with remote roles. **Adding AI/LLM projects + one advanced cert = ₹65-85L**. That's your lever.

Part 2: Market Context (2025 Reality)

Salary Benchmarks for Your Target Roles

Role	India (On-Site)	India (Remote/USD)	Global (USA/Canada)
Senior Data Engineer	₹35–50L	₹50–70L	\$120K–150K
Data/Cloud Architect	₹40–60L	₹60–90L	\$150K–180K
Senior AI/ML Data Engineer	₹45–70L	₹70–100L+	\$150K–200K
MLOps Engineer (Specialist)	₹40–65L	₹65–95L	\$140K–190K

Role	India (On-Site)	India (Remote/USD)	Global (USA/Canada)
Generative AI Engineer	₹50–80L	₹80–120L+	\$160K–250K

Key Insight: Your target—**₹70L+**—aligns with:

- **Remote USD-equivalent roles** (multinational or US startups)
- **Senior architect/leadership** positions at top Indian tech firms
- **Specialist AI/Data roles** blending engineering + ML

You're competing against people with **8-12 years + visible AI projects**. Your edge: **13 years + new AI expertise = strategic advantage**.

Market Demand (2025)

- **AI/Cloud roles growing 75% YoY** in India
- **MLOps roles grew 9.8× in 5 years** (LinkedIn)
- **Compensation for ML/MLOps jumped ~20% YoY**
- **India needs 1M AI professionals by 2026**—massive supply shortage
- **Generative AI Engineers command \$128K–150K globally** (~₹1.0–1.2Cr); even entry-level GenAI roles in India fetch ₹20–50L

Your Timing: Perfect. Act in the next 6 months before the wave saturates.

Part 3: 12-Month Execution Plan

Timeline Overview

Month 1–3: Phase 1 – AI/ML Foundations + First Cert
Month 4–6: Phase 2 – Build AI Projects + GitHub Visibility
Month 7–9: Phase 3 – Advanced Specialization + Networking
Month 10–12: Phase 4 – Job-Ready Positioning + Applications

Phase 1: AI/ML Foundations & Certification (Months 1–3)

Goal: Build ML/AI fundamentals, get first cloud cert, demonstrate commitment.

1.1 Core ML Learning (6–8 weeks, 10–15 hrs/week)

Why: You need hands-on understanding of ML pipelines, not just theory.

Courses & Resources:

a) **Andrew Ng's Machine Learning Specialization** (Coursera)

- Duration: 3–4 months at 5 hrs/week

- Focus: Supervised learning, neural networks, practical ML mindset
- Cost: ~₹2000–3000 (audit) or ₹20K (certificate)
- Output: Practical Python + scikit-learn projects

b) **Fast.ai – Practical Deep Learning** (Free)

- Focus: PyTorch, transfer learning, modern techniques
- Duration: 2–3 weeks (watch + code labs)
- Advantage: Industry-first approach; code-first learning

c) **LLM Essentials** (via Hugging Face tutorials + LangChain docs)

- Duration: 3–4 weeks self-study
- Topics: Transformer architecture, prompt engineering, vector DBs, LLM Ops basics
- Hands-On: Build a simple chatbot with Hugging Face + LangChain

Deliverable by End of Week 8:

- Completed Andrew Ng course (3 certificates)
- 2–3 mini-projects: e.g., (1) iris classification, (2) recommendation system, (3) simple sentiment classifier
- **GitHub repo** with code + README (signals action)

1.2 Cloud AI Platform Exploration (4 weeks, 5 hrs/week)

Why: You know BigQuery; now show you understand ML on cloud.

Deep Dives:

a) **GCP Vertex AI** (primary focus—aligns with your BigQuery expertise)

- AutoML basics (no-code/low-code ML)
- Custom training with Vertex Pipelines
- Model deployment and serving
- Hands-on: Train a simple model on BigQuery data → deploy via Vertex AI
- Codelab: <https://www.cloudskillsboost.google/>

b) **AWS SageMaker** (breadth—show versatility)

- Feature Store, Model Registry
- Basic pipeline setup
- Hands-on: Kaggle dataset → SageMaker notebook → model endpoint

Deliverable by End of Week 12:

- 1–2 Vertex AI projects (documented on GitHub or blog)
- Hands-on experience with model training, validation, and deployment

- Understanding of cloud ML platforms vs. on-prem

1.3 First Certification: Google Cloud Data Engineer (Week 8–12)

Why: Validates baseline cloud/data skills. Gateway to MLOps certs.

Plan:

- **Exam:** Google Cloud Professional Data Engineer
- **Prep Time:** 4 weeks (part-time)
- **Resources:**
 - Coursera: "Preparing for Google Cloud Data Engineer" (~2 weeks)
 - Linux Academy + Udemy practice exams (~2 weeks)
 - Official Google docs & whitepapers
- **Cost:** ~₹5K exam fee + ~₹15K course
- **Pass Rate Expectation:** ~70–80% (you already know most of this)

Why This First? It's your baseline. You'll ace it, it's on your resume, and it opens doors to AI/ML-focused certs.

Deliverable: GCP Data Engineer certification (expires 3 years)

Phase 1 Outputs

Item	Timeline	Evidence
ML foundations	Week 8	Coursera certs + GitHub mini-projects
Cloud AI experience	Week 12	Vertex AI / SageMaker hands-on projects
GCP Data Engineer cert	Week 12	Official cert badge
GitHub profile started	Week 12	3–4 repos with README, clean code

Phase 2: AI Project Development & GitHub Visibility (Months 4–6)

Goal: Build 2–3 production-quality AI projects blending your data expertise with ML/LLM.

2.1 Project 1: End-to-End ML Pipeline on GCP (Weeks 13–20, ~8 weeks)

Project: Customer Churn Prediction with Real-Time Scoring

Narrative: "Designed an enterprise-grade ML pipeline for predictive analytics, integrating large-scale data from BigQuery, training a classification model, and serving real-time predictions via a REST API."

Architecture:

1. **Data Ingestion & Transformation** (Your Strength)
 - Use Pub/Sub → BigQuery for streaming customer behavioral data

- Implement data validation, quality checks (like your PayPal work)
- Design bronze → silver → gold layer (medallion architecture)

2. Feature Engineering (New Skill)

- SQL/Pandas-based features from transaction data
- Time-based features (recency, frequency, monetary)
- Dimensionality reduction techniques

3. Model Training (ML Layer)

- LightGBM or XGBoost classifier (scikit-learn)
- Hyperparameter tuning with cross-validation
- Log experiments via MLflow

4. Deployment & Serving (MLOps Layer)

- Container the model (Docker)
- Deploy via Vertex AI Endpoints or Cloud Run
- Build a simple Flask REST API for predictions
- Set up CI/CD with GitHub Actions

5. Monitoring (Production Concerns)

- Track prediction latency, model drift
- Log predictions + actual labels (for model retraining)

Tech Stack: Python, Spark/SQL, BigQuery, Vertex AI, Docker, Flask, MLflow, GitHub Actions

Effort: ~40–50 hrs over 8 weeks

Outcome:

- End-to-end project (data → model → API)
- GitHub repo with clear documentation
- Blog post: "Building a Production ML Pipeline on GCP" (SEO + LinkedIn boost)

2.2 Project 2: Generative AI Application with LLM (Weeks 21–28, ~8 weeks)

Project: Intelligent Data Documentation Generator

Narrative: "Built a generative AI tool that automatically documents data schemas and pipelines using LLMs, integrating with enterprise knowledge systems via vector search."

Why This? Demonstrates **LLM + data engineering fusion**—exactly what ₹70L+ roles want.

Architecture:

1. Data Collection & Preparation

- Scrape/collect your own data pipelines' metadata (schemas, lineage, etc.)
- Build a knowledge base of pipeline documentation

- Organize in a structured format (JSON/Parquet)

2. **Vector Database Setup** (New Concept)

- Use Pinecone or Weaviate to index documentation
- Embed using OpenAI embeddings or HuggingFace models
- Practice semantic search

3. **LLM Integration**

- Use GPT-4 API (or open-source LLaMA via Hugging Face) for generation
- Build prompt templates for different data scenarios
- Implement Retrieval-Augmented Generation (RAG):
 - Query vector DB → retrieve relevant docs → feed into LLM → generate response

4. **Application Layer**

- Streamlit or Gradio app for user interaction
- Backend: FastAPI or Flask
- Deploy on GCP App Engine or Cloud Run

5. **Containerization & CI/CD**

- Docker containerization
- GitHub Actions for testing/deployment
- Monitoring: Log latency, token usage, errors

Tech Stack: Python, LangChain, OpenAI/HuggingFace Transformers, Pinecone/Weaviate, FastAPI, Docker, Streamlit

Effort: ~50–60 hrs over 8 weeks

Outcome:

- Working generative AI demo (live on a URL)
- GitHub repo + detailed README
- Blog: "Building RAG Systems for Enterprise Data Documentation" (₹70L+ hiring managers read this)
- Optional: Quick demo video (3–5 min) on LinkedIn

Why Powerful: You've built a **bridge** between your data expertise and modern AI. This is *exactly* what makes you hireable at ₹70L+.

2.3 **Docker & Kubernetes Basics (Concurrent, Weeks 13–28)**

Quick Skill Add (~2–3 hrs/week, 12 weeks)

- Docker fundamentals: containerize the above two projects
- Kubernetes basics: understand pod, deployment, service (not deep, just conceptual)
- Purpose: Show DevOps/DataOps maturity (expected for architect-level roles)

Resource: Freecodecamp Docker + Kubernetes tutorials (~8 hrs total)

Deliverable: Both projects containerized and deployable via `docker run` / `kubectl`

2.4 GitHub Profile & LinkedIn Visibility

GitHub:

- Create 3–4 polished repos (Projects 1, 2, and mini-projects from Phase 1)
- README with:
 - Problem statement & solution
 - Architecture diagram (use Excalidraw or Miro)
 - Tech stack
 - Instructions to run locally
 - Screenshots/results
- Clean code: proper variable names, docstrings, requirements.txt, .gitignore
- Regular commits (show iterative development, not one dump)

LinkedIn:

- Update profile headline: **"Senior Data Engineer | Cloud Data Architect | GenAI/LLM Integration"**
- Add projects section (link GitHub repos)
- Write 2–3 posts:
 - Week 16: "Why Every Data Engineer Needs ML Basics in 2025"
 - Week 20: "Building ML Pipelines on GCP: A Data Engineer's Guide"
 - Week 24: "Generative AI for Data Teams: A Practical RAG Example"
- Engage: Comment on relevant posts, follow AI/data thought leaders

Phase 2 Outputs

Item	Timeline	Evidence
Project 1 (ML Pipeline)	Week 20	GitHub repo + blog post
Project 2 (GenAI/RAG)	Week 28	Live demo + GitHub repo + video
Docker/Kubernetes basics	Week 28	Both projects containerized
GitHub profile polished	Week 28	3–4 public repos, clean code
LinkedIn visibility	Week 28	3 posts + updated profile

Phase 3: Advanced Specialization & Networking (Months 7–9)

Goal: Deepen expertise, network strategically, and position as "thought leader lite."

3.1 Second Certification: GCP Machine Learning Engineer or AWS Certified ML (Weeks 29–36, ~8 weeks)

Option A: GCP Machine Learning Engineer (Recommended—builds on Phase 1)

- Covers Vertex AI, model training, serving, monitoring
- Exam focus: ML workflows on GCP
- Prep: 4 weeks part-time
- Cost: ₹5K exam + ₹10K course materials

Option B: AWS Certified Machine Learning – Specialty (If you want AWS depth)

- Broader ML coverage
- Includes SageMaker, feature stores, etc.
- Prep: 4 weeks part-time

Pick One (Not Both Yet). GCP aligns better with your BigQuery expertise.

Deliverable: Second certification

3.2 Advanced Topics Deep Dive (Concurrent, 6–8 weeks)

Pick **one** based on interest:

a) MLOps/LLMOps Specialization

- Focus on model monitoring, drift detection, retraining pipelines
- Tools: MLflow, Weights & Biases, DVC
- Project: Set up automated retraining pipeline for Project 1
- Why: MLOps Engineers now command ₹65–95L (remote)

b) Feature Engineering & Feature Stores

- Deep dive into Feast or Tecton
- Design a production feature store for one of your projects
- Why: Differentiator for architects

c) Data Mesh & Lakehouse Architectures

- Study Databricks lakehouse concepts
- Design a multi-team data architecture (theoretical)
- Why: Architect-level thinking

Recommendation: Start with **MLOps** (most immediately valuable). You can rotate to others later.

Deliverable: Blog post or GitHub project demonstrating expertise

3.3 Networking & Community Engagement

Goal: Be visible, build relationships, and learn about ₹70L+ opportunities.

a) LinkedIn Actively

- Write 1–2 more posts on advanced topics (MLOps, LLMs, etc.)
- Engage: Reply thoughtfully to 5–10 posts/week from target companies' tech leaders
- Follow: Hiring managers at Google, Amazon, Microsoft, Databricks, etc.

b) Twitter/X (Optional but Effective)

- Share weekly learnings on AI/data/ML
- Retweet interesting papers, projects, insights
- Build a 500–1000 follower base of tech professionals

c) Open Source Contribution (2–4 weeks effort)

- Contribute to a popular repo: LangChain, Hugging Face, MLflow, or Feast
- Fix a bug or add a feature
- Even one merged PR signals credibility

d) Tech Talks & Meetups

- Attend or present at local data/AI meetups (Bangalore, Hyderabad, remote)
- Speak about: "Scaling ML Pipelines on GCP" or "LLMOps in Practice"
- Why: Builds visibility + confidence for senior interviews

e) Informational Interviews (Start in Month 8)

- Reach out to 10–15 Cloud Data Architects/Senior MLOps Engineers at your target companies
- Ask: 15-min call to discuss their journey, current work, skills gap
- Goal: Relationship + insider info on hiring, comp, requirements

3.4 Upskill in Sales/Communication (Soft Skill)

Why Important: At ₹70L+ roles, you need to **articulate business value**, not just technical depth.

- Read: "Cracking the PM Interview" (transposed for tech roles) or "Inspired" by Marty Cagan
- Practice: Craft 2–3 "elevator pitches" for your projects that lead with **business outcome**, not tech
 - Example: "Reduced customer churn prediction latency from batch (daily) to real-time (sub-second), enabling personalized interventions and a projected 15% retention improvement"
- Record yourself and review

Phase 3 Outputs

Item	Timeline	Evidence
Second cert (ML/GCP)	Week 36	Cert badge
Advanced specialization	Week 36	Blog/GitHub project
Open-source contribution	Week 34	Merged PR
Tech talk	Week 32	Recording or event link
10–15 informational interviews	Week 36	Relationships + intel
Communication practice	Week 36	3 refined pitches

Phase 4: Job-Ready Positioning & Applications (Months 10–12)

Goal: Final positioning, targeted applications, interview prep, and negotiation.

4.1 Resume & Profile Overhaul (Week 37–39)

Resume Changes:

- **Headline:** "Senior Data Engineer | Cloud Data Architect | Generative AI Integration"
- **Summary (2–3 lines):** Highlight the **AI/ML pivot** and breadth:
 - "Architected large-scale ETL pipelines (Spark, Kafka, BigQuery) and spearheaded AI/ML integration with LLM-based automation and real-time serving. Expertise in cloud architectures (GCP, Snowflake, Databricks) and MLOps workflows. 13+ years in data engineering; 2 cloud certifications; hands-on with GenAI tooling."
- **Skills Section:** Reorder by relevance for ₹70L+ role:
 - **Tier 1 (Highlight):** Cloud Architectures (GCP, AWS, Azure), ML/MLOps (Vertex AI, SageMaker, Kubeflow), Generative AI (LLMs, LangChain, Prompt Eng), Big Data (Spark, Kafka, Hive)
 - **Tier 2:** ETL/Data Pipelines, Data Warehousing (BigQuery, Snowflake, Databricks), DevOps/DataOps (Docker, Kubernetes, CI/CD)
 - **Tier 3:** Programming (Python, Scala, Java, SQL), Databases (Oracle, MySQL, NoSQL)
- **Experience Section:**
 - Current role (Home Depot): Bullet 1 = architecture decision, Bullet 2 = ML/AI impact, Bullet 3 = mentorship
 - Example: "Led architecture and implementation of ML-enabled ETL pipeline using Apache Airflow + Vertex AI, enabling real-time fraud detection and reducing false positives by 25%"
 - Emphasize: scale (data volume), business impact (revenue/cost/efficiency), leadership
- **New Sections:**
 - **Projects:** Add links to GitHub Projects 1 & 2 with 1-line descriptions

- **Certifications:** GCP Data Engineer, GCP ML Engineer, etc.
- **Publications/Thought Leadership:** Links to your 3–4 LinkedIn posts or blog

LinkedIn Profile:

- **Headline:** Same as resume
- **About:** 3–4 sentences + emojis for readability
- **Featured:** Pin your 2 best projects + blog posts
- **Background:** Update current role description to highlight AI/ML; add projects section
- **Recommendations:** Ask 3–5 people for recommendations (if you have strong relationships)

4.2 Identify Target Companies & Roles (Week 37–40)

Strategy: Cast a wide net but with precision.

Tier 1 (Dream Companies—₹70–90L+):

- Google India (Cloud/Data roles)
- Amazon India (AWS, data/AI)
- Microsoft India (Azure, Fabric)
- Databricks (any region, remote)
- Stripe, Figma, [Checkout.com](https://www.checkout.com/) (remote-first, USD)
- Startups: NVIDIA, Anthropic, etc. (remote, stock options)

Tier 2 (Solid Alternatives—₹60–75L):

- TCS, Infosys, Accenture (India, leadership track)
- Unicorn startups: Unacademy, Razorpay, PharmEasy, etc.
- Consulting: Deloitte, EY, McKinsey (data/AI practices)

Tier 3 (Backup—₹50–65L):

- Regional tech firms, financial services

Roles to Target:

- "Senior Data Engineer (Cloud/AI)"
- "Cloud Data Architect"
- "Senior MLOps Engineer"
- "AI/Data Solution Architect"
- "Head of Data Engineering" (if leadership track is goal)

Effort: Shortlist 20–30 roles. Track in a spreadsheet.

4.3 Targeted Applications (Week 40–48, ongoing)

Apply to:

- **Tier 1 roles:** 5–7 per week (quality over quantity)
- **Tier 2 roles:** 3–5 per week
- **Tier 3 roles:** 2–3 per week (only if Tier 1/2 stall)

Application Strategy:

- Tailor cover letter (500 words):
 - Highlight **1 project** that aligns with the company's tech stack
 - Mention a **specific technology** from the job post you've used
 - Example: "Your data platform's use of Databricks aligns with my recent ML pipeline project..."
- Personalization: Research hiring manager on LinkedIn; if possible, reference their recent talk/post
- **Leverage Networks:**
 - Ask your informational interview contacts if they can refer you (hugely increases callback rate)
 - Post on LinkedIn: "Open to exploring Senior Data Architect / MLOps roles, especially at [Companies]" (get referrals)

Expected Callback Rate:

- Generic application: ~5–10%
- Tailored cover letter: ~15–20%
- With referral: ~30–50%

Goal: 3–5 interviews lined up by Week 45

4.4 Interview Preparation (Week 40–48, concurrent)

Prepare for 3 Rounds:

Round 1: Technical Screening (30–45 min)

- Problem: System design for ML pipeline or data architecture
- Expect: "Design a real-time churn prediction system for an e-commerce platform"
- Prep:
 - Study: "Designing Machine Learning Systems" (book or free resources)
 - Mock interviews: InterviewBit, LeetCode, System Design Primer
 - Practice: 3–5 mock system-design interviews

Round 2: Deep Dive / Project Discussion (45–60 min)

- Discussion of your GitHub projects

- Expect: "Walk us through your RAG system. What alternatives did you consider? Why did you choose Pinecone over Weaviate?"
- Prep:
 - Be ready to discuss: architecture trade-offs, tech choices, lessons learned, scalability concerns
 - Practice with friend/mentor
 - Record yourself explaining Project 1 & 2 (audio, 5 min each)

Round 3: Leadership & Behavioral (30–45 min)

- STAR framework questions: "Tell me about a time you led a difficult technical decision"
- Situational: "How would you mentor a junior engineer struggling with cloud architecture?"
- Prep:
 - Craft 5–6 STAR stories tied to:
 - Mentorship (you have this from PayPal/Home Depot roles)
 - Technical leadership (choosing Spark over MapReduce, migrating to GCP)
 - Handling conflict/ambiguity
 - Cross-team collaboration
 - Practice with mock interviewer

Negotiation Prep:

- Research comp using Levels.fyi, Blind, Glassdoor
- Walk-in number: ₹70L base + ₹10L bonus + stock = ₹80L
- Be ready to say: "I'm open to ₹70–85L based on equity and role scope"
- Don't commit early; let them make the first offer

4.5 Continuous Learning (Week 40–52, 3–5 hrs/week)

- **New Trends:** Subscribe to Towards Data Science, AI news (ThePrdctctiveAndrew, same for new papers)
- **Tool Exploration:** Try 1 new tool/framework per month (e.g., LLaMA, Ollama, Ray, Argo Workflows)
- **Reading:** 1 technical book (e.g., "Building Machine Learning Systems at Scale" or "LLM Engineering")

Phase 4 Outputs

Item	Timeline	Evidence
Resume overhaul	Week 39	Updated, links included
LinkedIn profile	Week 39	Optimized, pinned projects

Item	Timeline	Evidence
Target list (20–30 roles)	Week 40	Spreadsheet + tracking
Applications sent	Week 48	60–80 applications
Interviews lined up	Week 45	3–5 phone screens
Interview prep	Week 48	Mock interviews done
Offers received & negotiated	Week 52	₹70L+ package

Part 4: Roadmap Overview (Visual Timeline)

PHASE 1: Foundations (Weeks 1–12)		
└ ML courses (Coursera, Fast.ai)		→ Weeks 1–8
└ Cloud AI platforms (Vertex AI, SageMaker)		→ Weeks 5–12
└ GCP Data Engineer cert		→ Weeks 8–12
└ GitHub setup + mini-projects		→ Weeks 1–12
PHASE 2: AI Projects (Weeks 13–28)		
└ Project 1: ML Pipeline (BigQuery → Model → API)		→ Weeks 13–20
└ Project 2: GenAI/RAG (Chatbot + Vector DB)		→ Weeks 21–28
└ Docker/Kubernetes basics		→ Weeks 13–28
└ GitHub repo polishing		→ Weeks 15–28
└ LinkedIn posts + visibility		→ Weeks 16–28
PHASE 3: Specialization (Weeks 29–36)		
└ GCP ML Engineer cert (or AWS ML specialty)		→ Weeks 29–36
└ Advanced topic deep dive (MLOps preferred)		→ Weeks 29–36
└ Open-source contribution		→ Weeks 29–34
└ Tech talk/meetup presentation		→ Weeks 29–34
└ Informational interviews (10–15 people)		→ Weeks 32–36
PHASE 4: Market Ready (Weeks 37–52)		
└ Resume & profile overhaul		→ Weeks 37–39
└ Target company list (20–30)		→ Weeks 37–40
└ Job applications (60–80 total)		→ Weeks 40–48
└ Interview prep & mocks		→ Weeks 40–48
└ Offers negotiation		→ Weeks 48–52
GOAL: ₹70L+ Offer by Week 52		

Part 5: Estimated Effort & Investment

Time Commitment

Phase	Weekly Hours	Total Hours	Notes
Phase 1	15	180	Intense; courses + certification
Phase 2	12	96	Project development, concurrent work

Phase	Weekly Hours	Total Hours	Notes
Phase 3	10	30	Lighter; working full-time + side work
Phase 4	8	32	Job hunt + interview prep (part of normal job search)
TOTAL	11–15 avg	~338 hours	~6 months solid effort + 6 months lighter

Reality Check: This is aggressive but doable alongside a full-time job. If you're disciplined (5 hrs/week × 52 weeks), it's manageable. Weekends + holidays are your friends.

Financial Investment

Item	Cost	Justification
Coursera (ML + Cloud)	₹30K	Audit free, but cert costs extra
Google Cloud credits (free tier)	₹0	Google gives ₹12K/month free for new accounts
AWS/GCP certifications (2)	₹10K	Exam fees
Udemy/Linux Academy courses	₹5K	Practice exams + additional content
GitHub Pro (optional)	₹0	Free for you; standard accounts are fine
Domain + blog (optional)	₹2K	sivarajmaladi.dev or similar; boosts credibility
TOTAL	~₹47K	ROI: ₹70L offer = 1500× return

Optional Investments:

- 1-on-1 mentoring from a Cloud/AI architect: ₹5–10K for 5–10 sessions (accelerates learning)
- Conference/workshop attendance: ₹5–15K (networking + credibility)

Part 6: Success Metrics & Checkpoints

Monthly Checkpoints

Month	Key Deliverables	Status Check
M1	Coursera course started, mini-projects 1–2 done	2 GitHub repos, 20 hrs effort
M2	Coursera finished, GCP cert scheduled	1 cert badge, 4 repos
M3	GCP cert earned, Vertex AI project live	Cert + 1 production project
M4	ML pipeline project halfway, 1 blog post	GitHub active, LinkedIn posts
M5	ML pipeline + RAG project done, GitHub polished	2 production projects, 500+ GitHub stars?
M6	3–4 LinkedIn posts, Docker/K8s basics, informational interviews started	Visible community presence
M7	2nd cert earned, MLOps deep dive, open-source PR merged	Advanced specialization visible

Month	Key Deliverables	Status Check
M8	Tech talk done, 10+ informational interviews	Network + credibility
M9	Advanced expertise blog posts, target company research	₹70L+ roles identified
M10	Resume + LinkedIn revamp, 30 applications sent	Applications tracked
M11	Interviews happening, mocks completed	3–5 interviews
M12	Offers negotiated, ₹70L+ target achieved	Celebration! 🎉

Success Criteria for Each Phase

Phase 1:

- ✓ GCP Data Engineer cert earned
- ✓ 3–4 GitHub repos with clean code and READMEs
- ✓ ML fundamentals understood (can explain neural networks, backprop)

Phase 2:

- ✓ 2 production-quality projects (not demos)
- ✓ Projects use modern stack (MLflow, Docker, cloud platforms)
- ✓ Each project has a blog post + LinkedIn visibility
- ✓ GitHub profile shows consistent activity

Phase 3:

- ✓ 2nd cert (GCP ML or AWS ML)
- ✓ 1 open-source contribution (merged PR)
- ✓ 1 tech talk or meetup presentation
- ✓ 10–15 informational interviews completed (relationships built)

Phase 4:

- ✓ 60–80 targeted applications
- ✓ 3–5 interviews reached
- ✓ ₹70L+ offer negotiated and accepted

Part 7: Key Recommendations & Considerations

1. Leverage Your Strengths

Trading Framework Interest → ML/Quantitative Finance Pivot (Optional)

You mentioned interest in trading frameworks. This is a *secondary* opportunity:

- **Quantitative traders/analysts** command ₹80L–₹1.5Cr+ globally
- **ML engineers in fintech** (prediction, risk models) fetch ₹60–80L+ in India
- **Timeline:** After ₹70L data role is achieved, pivot to fintech if interested
- **Resources:** Intro to quantitative trading (Coursera), build a simple backtesting engine

For Now: Keep it as a future specialization. Focus on general AI/data architect first.

2. Continuous Learning Framework

You mentioned "continuous learning" as a strength. **Institutionalize it:**

- Weekly: 1 new research paper or GitHub project summary (30 min)
- Monthly: 1 online course section or tutorial (4–6 hrs)
- Quarterly: 1 new tool or framework exploration (8–12 hrs)
- Yearly: 1 certification or advanced specialization

This keeps you ahead of the curve post-promotion.

3. Leverage AI for Learning (Your Stated Interest)

Use AI tools to **accelerate** your learning:

- ChatGPT + code generation: Use for boilerplate code, debugging, conceptual explanations
- GitHub Copilot: Write cleaner, faster code
- Claude: Deep technical research, explaining complex papers
- Perplexity/Google Gemini: Research current trends, find resources

Time Savings: ~30% faster learning with AI assistance. Reinvest that time into more projects.

4. Personal Brand Building

- **LinkedIn:** Post weekly (or bi-weekly). Consistency > frequency.
- **Blog:** Write 8–10 substantive posts over 12 months. Aim for 1000+ word deep dives.
- **GitHub:** Show your best work. Clean code matters more than quantity of projects.
- **Twitter/X:** Optional but effective for AI community. 3–5 posts/week.

ROI: Strong personal brand leads to inbound recruiter interest (highest pay negotiations).

5. Negotiation Strategy

Once you have an offer at ₹60–65L, **negotiate:**

- Baseline ask: ₹75L + ₹10L performance bonus + stock/ESOP
- If they counter with ₹70L + ₹5L bonus, accept (that's ₹75L total)
- Remote vs. on-site: Remote = 10–15% salary bump globally
- Equity/stock: In startups, negotiate for 0.1–0.5% if they offer it

Psychology: They called you; they want you. Be confident.

6. Risk Mitigation

- **What if job market slows?** You'll be qualified for ₹55–65L roles (still solid)
- **What if certifications don't help?** Projects + GitHub visibility still get you hired
- **What if projects fail?** Failure is learning. Blog about it. Show resilience.
- **Backup Plan:** If after 12 months you're at ₹60L, extend 6 more months for ₹70L roles

Part 8: Your Next 48 Hours

Immediate Actions (This Week)

1. **Enroll in Coursera:** Register for Andrew Ng's ML specialization. Spend 2 hrs on Week 1 lectures.
2. **Set Up GitHub:** Create a clean GitHub profile (if not done), update bio to "Senior Data Engineer | Learning AI/ML"
3. **Google Cloud Credits:** Sign up for \$300 free credits (<https://cloud.google.com/free>). Play with Vertex AI for 1 hour.
4. **Schedule Cert Study:** Block 4–5 hrs/week for the next 12 weeks for GCP Data Engineer cert. Add to calendar.
5. **Find 1 Mentor (Optional):** Reach out to 1 person (senior architect, senior engineer at target company) for a 15-min call. Ask for guidance on the plan.
6. **Create Spreadsheet:** Track your progress—time spent, resources consumed, projects completed. Review weekly.

Part 9: Final Word

You are positioned for a significant leap—from ₹50–60L to ₹70–80L+. The gap is not talent or experience; it's **strategic positioning**.

The market is screaming for your profile: **senior data engineers with AI/ML chops, cloud expertise, and leadership.**

6–9 months of focused effort + visibility = ₹70L+ package.

Your current role at Home Depot is excellent for your foundation. Use the next 6 months to **build, learn, showcase, and network**. By Month 9–10, you'll be job-ready. By Month 12, you'll have offers.

Key Mindset: This is not about working *harder*; it's about working *smarter*—learning the right skills, building visible proof, and letting the market reward you.

Appendix: Resource Links

Courses & Learning

- Andrew Ng ML Specialization: <https://www.coursera.org/specializations/machine-learning-introduction>
- Fast.ai: <https://www.fast.ai/>
- Google Cloud Training: <https://cloud.google.com/training>
- LangChain Docs: <https://python.langchain.com/>
- Hugging Face Course: <https://huggingface.co/course>

Certifications

- GCP Data Engineer: <https://cloud.google.com/certification/data-engineer>
- GCP ML Engineer: <https://cloud.google.com/certification/machine-learning-engineer>
- AWS ML Specialty: <https://aws.amazon.com/certification/certified-machine-learning-specialty/>

Tools & Frameworks

- MLflow: <https://mlflow.org/>
- Vertex AI: <https://cloud.google.com/vertex-ai>
- LangChain: <https://python.langchain.com/>
- Streamlit: <https://streamlit.io/>
- Docker: <https://www.docker.com/>
- Kubernetes: <https://kubernetes.io/>

Community & Networking

- Kaggle: <https://www.kaggle.com/>
- GitHub: <https://github.com/>
- LinkedIn: <https://www.linkedin.com/>
- Towards Data Science (Medium): <https://towardsdatascience.com/>
- Local Meetups: [Meetup.com](https://www.meetup.com/) (search "data engineering," "AI," "machine learning" in your city)

Revision Date: November 2025

Status: Ready to Execute

Target Outcome: ₹70L+ Offer by November 2026

[1] [2] [3] [4] [5] [6] [7] [8] [9] [10] [11] [12] [13] [14] [15] [16] [17] [18] [19] [20] [21]

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3. <https://www.learnovita.com/cloud-architect-salary-in-india-article>
4. [https://www.ultratechcement.com/content/dam/ultratechcementwebsite/pdf/financials/annual-reports/Integrated-and-Sustainability-Report-FY24-\(double-page\).pdf](https://www.ultratechcement.com/content/dam/ultratechcementwebsite/pdf/financials/annual-reports/Integrated-and-Sustainability-Report-FY24-(double-page).pdf)
5. <https://dailyremote.com/salaries/data-engineer>
6. <https://www.tredence.com/blog/generative-ai-llm-engineer-careers-2025>
7. <https://qubit-labs.com/ai-engineer-salary-guide/>
8. <https://www.peopleinai.com/blog/the-job-market-for-mlops-engineers-in-2025>
9. <https://futureense.com/blog/ai-skills-in-demand>
10. <https://careers.oerlikon.com/job/Brugherio-Senior-Cloud-Data-Architect-20861/1141271001/>
11. <https://www.indeed.com/career-advice/finding-a-job/how-to-become-quantitative-trader>
12. Industry-Trends-Role-Overview-Q4-2025.pdf
13. <https://acarasolutions.in/blog/the-growing-demand-for-tech-talent-in-india/>
14. <https://www.striim.com/blog/data-architect-vs-data-engineer-an-overview-of-two-in-demand-roles/>
15. Prompt.md
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